23AID205 - Introduction to AI and Machine Learning

Title: Healthcare Recommendation System

Presented by Team- A8

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INTRODUCTION

- The healthcare industry struggles to provide personalized treatment due to reliance on generalized advice.
- This excerpt discusses a system using machine learning to analyze various datasets, including symptoms, medications, diet plans, and workout regimes, to deliver tailored healthcare recommendations.
- The goal is to enhance patient health outcomes by addressing the gap in individualized care.

OBJECTIVES

01. To develop a machine learning model to predict diseases based on patient symptoms, improving diagnostic accuracy.

02.To reduce the time taken for initial diagnosis by automating the symptom analysis process, streamlining patient care workflows.

03.To create a system that can be scaled and integrated into telemedicine platforms, enabling remote consultations and healthcare access to a wider audience.

PROBLEM STATEMENT

To develop a machine learning-based system that predicts diseases based on user-input symptoms. The system will provide additional information such as disease descriptions, medications, precautions, and diet recommendations and make accurate predictions through a well-trained model.

METHODOLOGY

1. **Data Collection and Data Preprocessing:** Collecting data and Loading datasets related to symptoms, diet, medication, workout routines, precautions and medical descriptions. Cleaning the data by handling missing values, for each data frame filling unknown values. Encoding categorical variable and normalizing numerical features.
2. **Model Training:** Spliting the data into training and testing sets. Using ML models (Logistic Regression, Random Forest Classifier, XGBoosting, Neural Networks) to train the dataset and predict health recommendations.

METHODOLOGY

3. **Evaluation:** Evaluating the each model using metrics and taking the good accuracy model for further integration.

4. Hyper Parameter tuning the Random Forest through GridSearchCV to identify the best parameters.

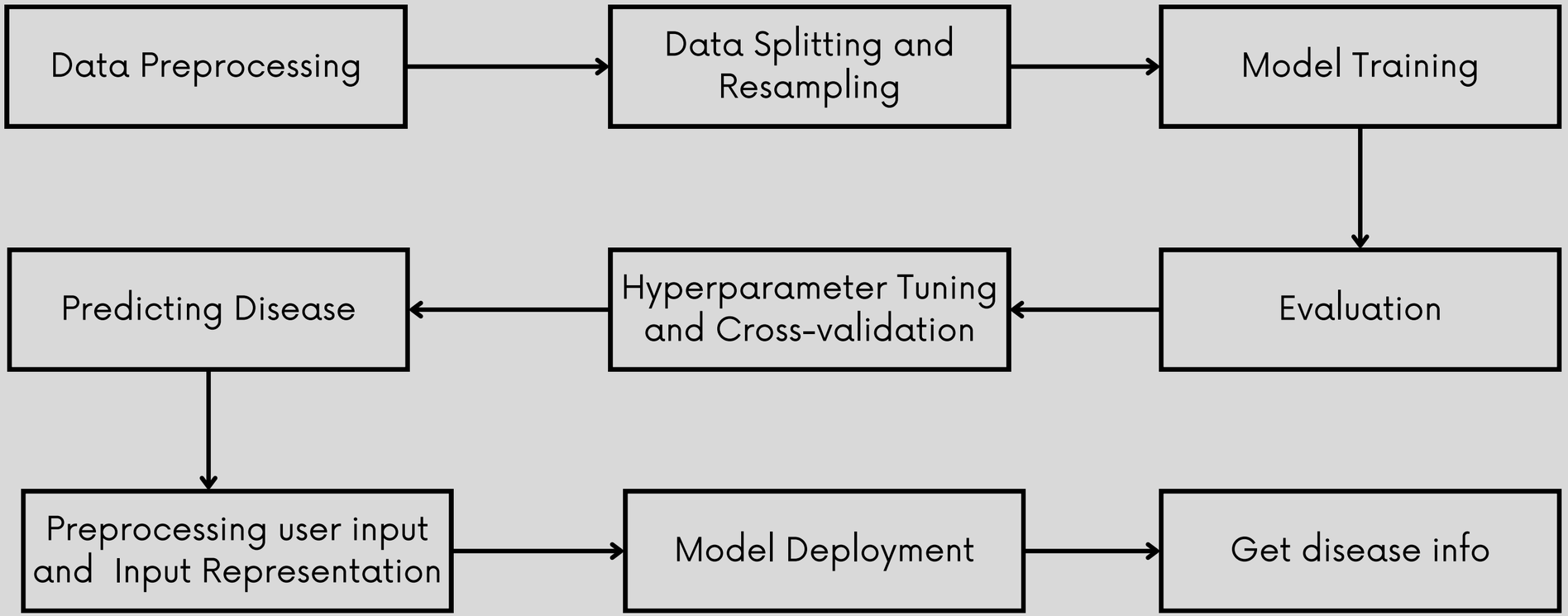
5. **Integration:** After training the model, integrating the recommendation to provide health suggestions based on patient data.

METHODOLOGY

Once the predicted disease is determined, the system fetches relevant medical information associated with the disease from the different datasets.
This includes:

- Description of the disease.
- Medications typically prescribed for the disease.
- Diet recommendations for the disease.
- Precautions to take.
- Workout recommendations

METHODOLOGY



METHODOLOGY

Random Forest is based on the idea that combining multiple decision trees can lead to more accurate predictions than a single decision tree. Each individual tree might be slightly different due to randomness, but when their predictions are combined, they form a more robust model.

Random Forest Hyperparameters:
Some important hyperparameters in Random Forest include:

- Number of trees (n_estimators): The number of decision trees to build. More trees usually lead to better performance but increase computation time.
- Maximum depth (max_depth): Limits the depth of each tree. Shallow trees reduce overfitting but may underfit the data.
- Minimum samples per split (min_samples_split): The minimum number of samples required to split an internal node.

DECISION TREE

It works by breaking down a dataset into smaller subsets while incrementally building an associated tree.
At each node of the tree, the data is split based on specific criteria, and this process continues until the subsets at the leaves become pure or meet certain stopping criteria.
Each node is basically a feature and we split the nodes based on a criteria

How to check if a split is pure split

- There are 2 metrics to check if a split is pure or not, Gini and Entropy.

$$Gini = 1 - \sum_{i=1}^{n} p_i^2$$

Where:

- p_i is the proportion (probability) of class i in the node.
- n is the total number of classes.

We prefer Gini over entropy because Gini is faster to compute as it involves only squaring probabilities, whereas entropy requires logarithmic calculations.



DECISION TREE

- Number of samples = 169.
- The value array: `[3.0, 8.0, 1.0, 11.0, 9.0, 14.0, 5.0, 6.0, 4.0, 1.0, 5.0, 10.0, 9.0, 10.0, 10.0, 5.0, 6.0, 4.0, 4.0, 4.0, 6.0, 8.0]`.

1. First, sum the values in the array:

   $$3+8+1+11+9+14+5+6+4+1+5+10+9+10+10+5+6+4+4+4+6+8=169$$

   So, total samples $P=169$.

2. Now calculate the proportion $\frac{p_i}{P}$ for each class:

   $$\frac{3}{169},\frac{8}{169},\frac{1}{169},\frac{11}{169},\frac{9}{169},\frac{14}{169},\cdots,\frac{8}{169}$$

3. Square these proportions:

   $$\left(\frac{3}{169}\right)^2=0.000314,\left(\frac{8}{169}\right)^2=0.00224,\left(\frac{1}{169}\right)^2=0.000035,\ldots$$

4. Sum these squared proportions:

   $$0.000314+0.00224+0.000035+\cdots+0.00224=0.030$$

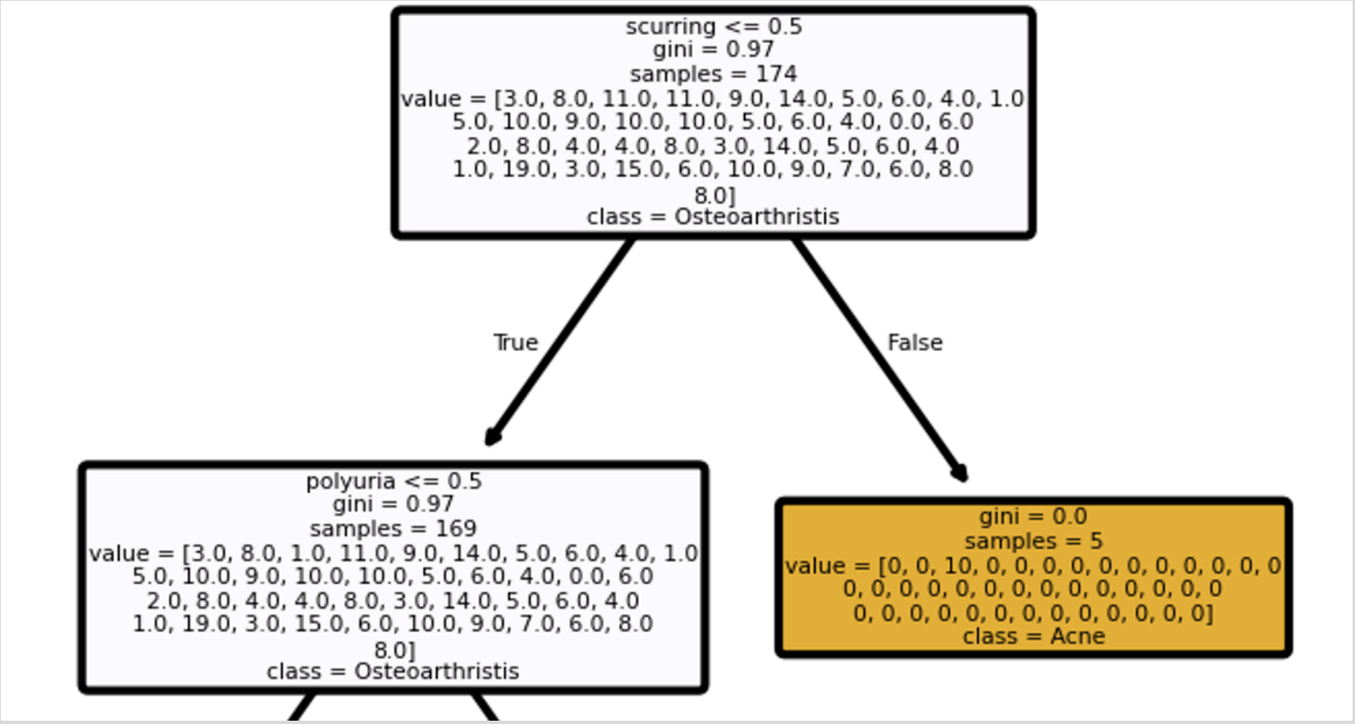
5. Calculate the Gini Index for the node:

   $$\text{Gini}=1-0.030=0.97$$

DECISION TREE

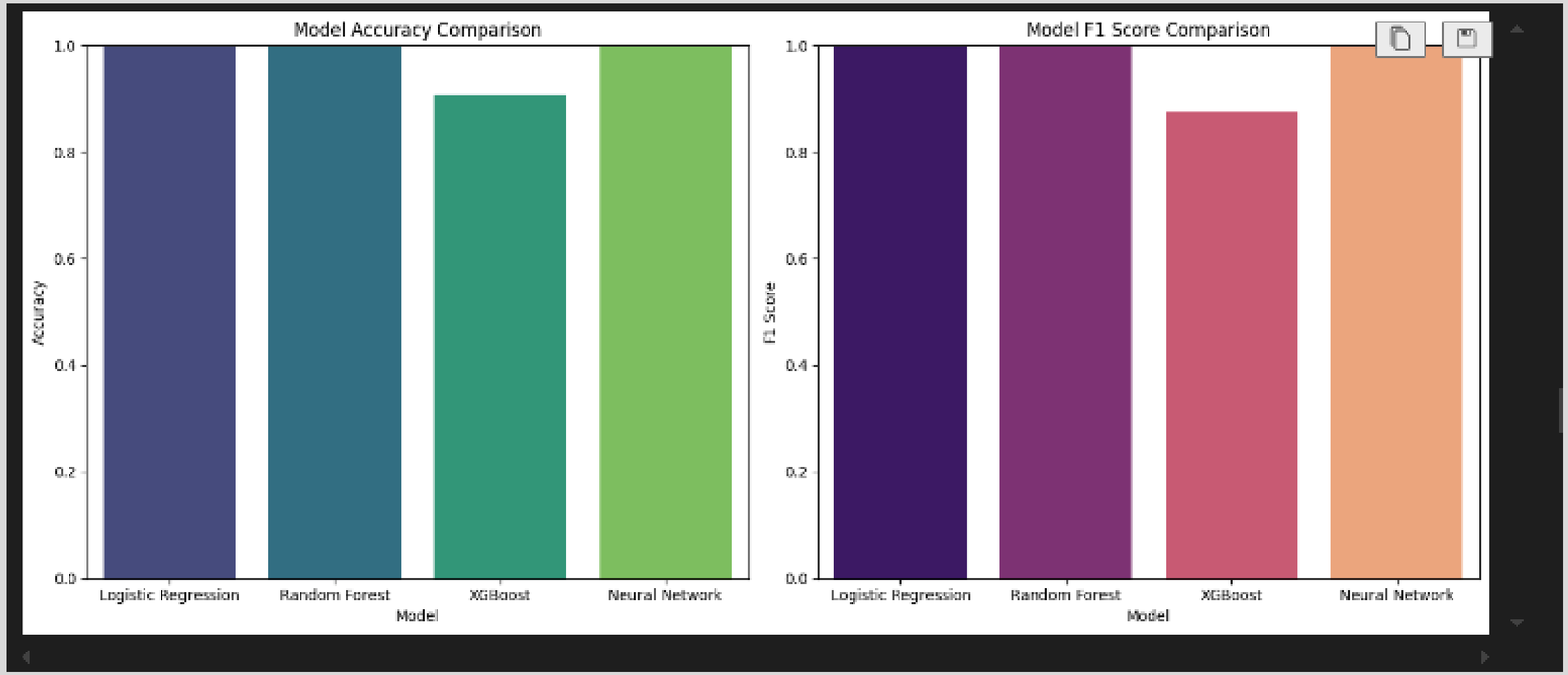
Ok, but why are the sum of values not equal to sample size??

In Random Forests, each tree is trained on a random subset of the training data, and this subset is created through sampling with replacement. This means that some data points may appear multiple times in the training set for a particular tree, while others may not appear at all.



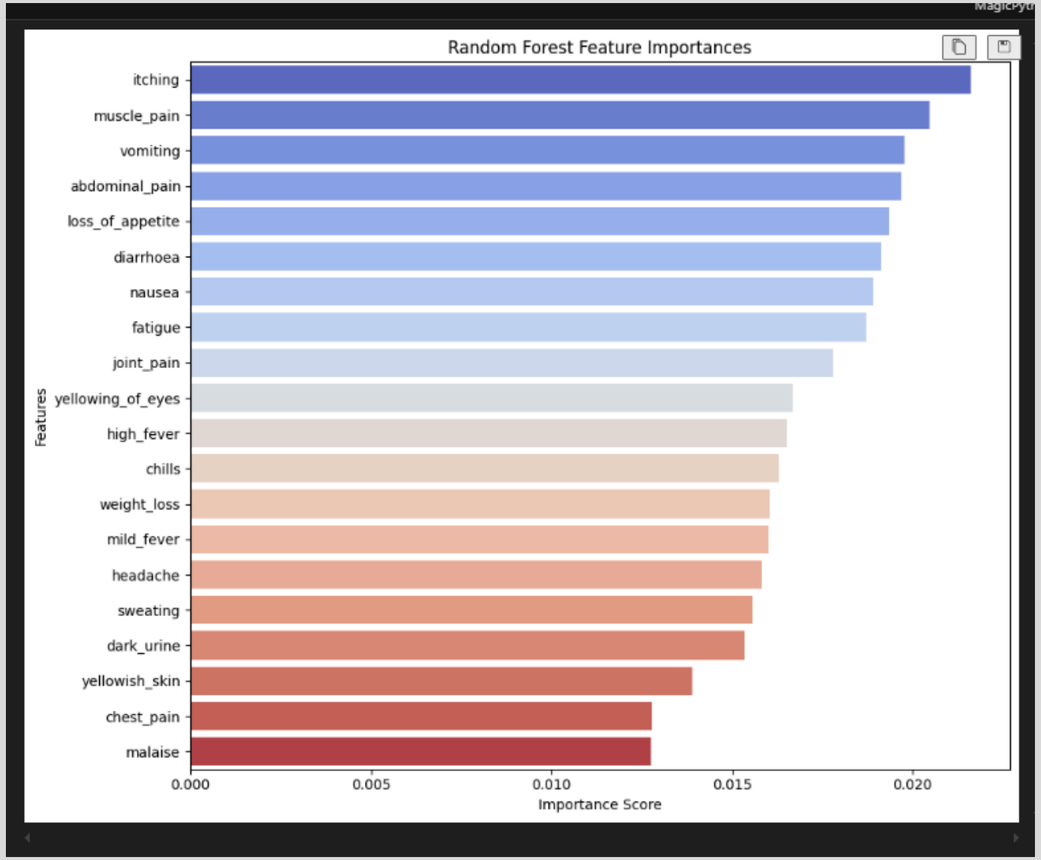
RESULTS

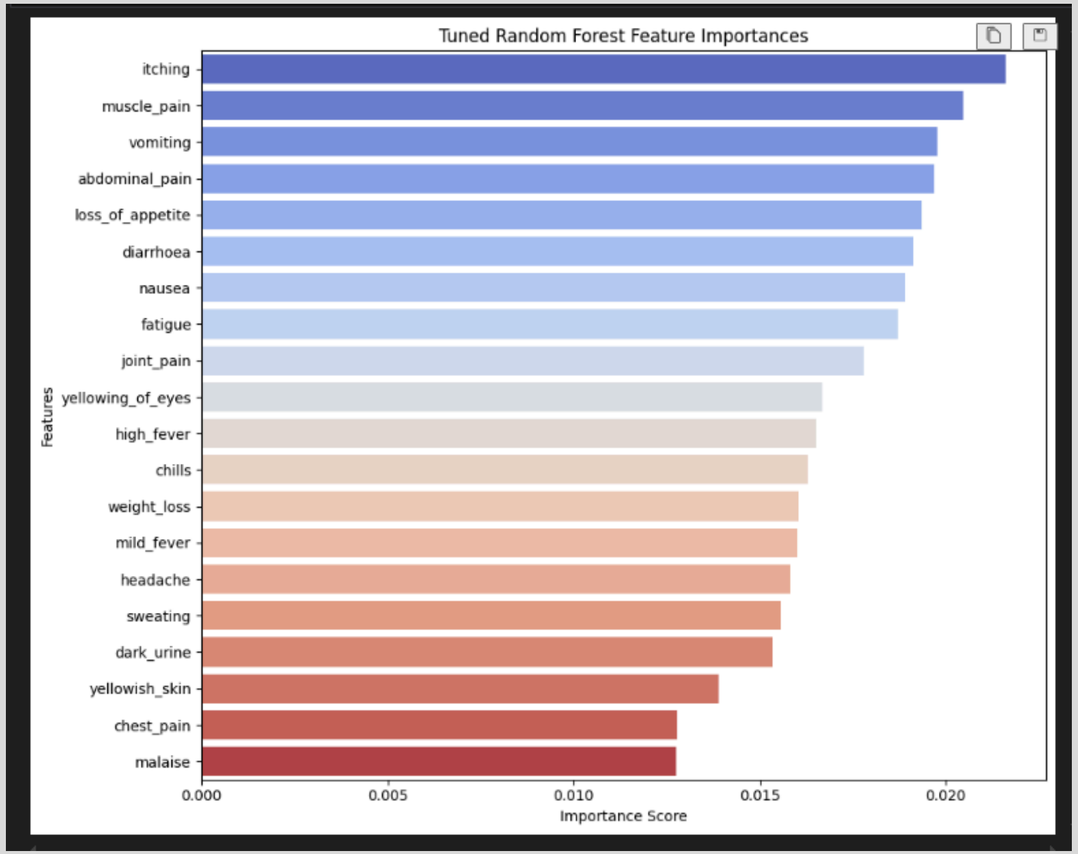
Model Comparison



RESULTS

Feature Importances





RESULTS

Final Output:
Predicted Disease with Medical info

```
================ Predicted Disease ================
Fungal infection

================ Description ================
Fungal infection is a common skin condition caused by fungi.

================ Precautions ================
1. bath twice
2. use detol or neem in bathing water
3. keep infected area dry
4. use clean cloths

================ Medications ================
1. ['Antifungal Cream', 'Fluconazole', 'Terbinafine', 'Clotrimazole', 'Ketoconazole']

================ Workouts ================
1. Avoid sugary foods
2. Consume probiotics
3. Increase intake of garlic
4. Include yogurt in diet
5. Limit processed foods
6. Stay hydrated
7. Consume green tea
8. Eat foods rich in zinc
...
10. Eat fruits and vegetables

================ Diets ================
1. ['Antifungal Diet', 'Probiotics', 'Garlic', 'Coconut oil', 'Turmeric']
```

RESULTS

```
Predicted Disease: Fungal infection
```

```
Description: Fungal infection is a common skin condition caused by fungi.
Medications: ['Antifungal Cream', 'Fluconazole', 'Terbinafine', 'Clotrimazole', 'Ketoconazole']
Diet Recommendations: ['Antifungal Diet', 'Probiotics', 'Garlic', 'Coconut oil', 'Turmeric']
Precautions: bath twice, use detol or neem in bathing water, keep infected area dry, use clean cloths
Workout Recommendations: Avoid sugary foods
```

CONCLUSION

In this project, we successfully developed a comprehensive healthcare recommendation system aimed at predicting diseases based on user-provided symptoms and delivering medical information, including medications, precautions, diets, and workout recommendations. The journey from data preprocessing to model deployment involved meticulous steps to ensure accuracy, reliability, and usability of the system.